



Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad Cantonment, Natore-6431

Faculty of Electrical and Computer Engineering

Department of Computer Science and Engineering

Lesson Plan

Program:	Computer Science and Engineering (CSE)
Course Code & Title:	CSE-3106 Operating Systems Sessional
Semester:	Winter-2025
Batch:	18th [Syllabus Version: 3]
Level:	3 rd Year 1 st Semester
Credit Hour:	1.50 Hr.
Curriculum Structure	Outcome Based Education (OBE)
Name & Designation of Teacher:	Subrata Kumer Paul Assistant Professor, Dept. of CSE Rakhi Rani Paul Lecturer, Dept. of CSE
Class Hours:	Tuesday: 01:35 PM – 02:35 PM Wednesday: 08:15 AM – 11:10 AM
Consultation Hours:	Sunday: 08:15 AM – 11:10 AM
E-mail:	sksubrata96@gmail.com (Subrata Kumer Paul) rakhipaul.cse@gmail.com (Rakhi Rani Paul)
Mobile:	+8801738061325 (Subrata Kumer Paul) +880174429743 (Rakhi Rani Paul)
Rationale:	The Operating System (OS) Sessional course is designed to provide hands on understanding on basic components of Operating Systems. The lab begins with the activities related to development of operating systems like UNIX and WINDOWS. Subsequently the course deals with virtualization and different key components of Operating System e.g., kernel compilation, process and thread scheduling, deadlocks, memory management, synchronization and system call etc.
Objective	1. To learn basic OS concepts and to be familiar with the design principles of Operating System 2. To know the internal and design principles of Operating System
Pre-requisite (if any):	None

Course Outcomes (CO):

Upon completion of the course, the student will be able to:

1.	Understand the basic OS concepts be familiar with the design principles of the Operating System and practice different Linux commands.
2.	Apply and implement different types of CPU scheduling algorithms for process management.
3.	Develop and design algorithms for deadlock and Disk scheduling algorithms.

Course Content:

- Introduction: Development of Linux Operating System, Installation of Linux in various modes, Installation of Windows application programs on Linux, Basic Linux Command
- Linux Kernels and Office Environments: Compilation
- Shell Programming: variables, statements, loop, array, functions, etc.
- Memory management: preemptive and non-preemptive algorithms and implementation
- Interposes communication and Process scheduling: algorithms and implementation
- Mutual exclusion and deadlock: algorithms and implementation
- Security of Windows and UNIX-like OS: hardening and security issues.

Text Book (s):

1. Abraham Silberschatz and Peter Baer Galvin, “Operating Systems Concepts”, Ninth edition, John Wiley & Sons, Inc., ISBN 978-1-118-06333-0

Reference Book(s):

1. Andrew S. Tanenbaum, “Modern Operating Systems” (4th edition) - Prentice Hall (2014), ISBN:978-0-13-600663-3
2. McGraw-Hill- Madnick and J. Donovan, Operating systems (1st edition), ISBN-10: 0070394555
3. Terrence Chan, Unix System Programming in C++ (1st edition), Prentice Hall Publication, ISBN-10: 0133315622

***Additional reading material:** PowerPoint slides, Docx, PDF, and web resources provided by the Teacher.

Assessment vs. CO/WA (PO)/Bloom’s Taxonomy Level/WP/EA Mapping **Detailed Mapping (According to BAETE Accreditation Manual- III)**

Mapping CO-PO, Bloom’s Level (BL), Knowledge Profile (WK), Complex Engineering Problem (WP) and Complex Engineering Activities (EA)

Course No & Title: CSE-3108 (Operating Systems Sessional)

COs	CO Statements	PO	WK	WP	EA	Bloom’s Level	Assessment Tools
CO1	Understand the basic OS concepts to be familiar with the design principles of the Operating System and practice different Linux commands.	1	2 4 5	1	1	C1, C2	Lab test/Project, Written test/Quiz, Viva Voce, Attendance, Data Sheet/Home Assignments/ Report, Class Performance /Continuous Evaluation)
CO2	Apply and implement different types of CPU and Disk scheduling algorithms for process management.	2				C3, C4	
CO3	Develop and design algorithms to address deadlock prevention, detection, and resolution, as well as page replacement techniques for memory management.	3				C4, C5	

Legends: CO: Course Outcomes PO: Program Outcomes, KP: Knowledge Profile

Bloom’s Level: Cognitive (C1-C6), Psychomotor(P1-P7), Affective(A1-A5)

CO-PO Skill Mapping

COs	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1	H											
CO2		H										
CO3			H									

Legends: High(H)=3, Medium(M)=2 and Low(L)=1

Teaching-learning Activities

- ☒ Lecture
- ☒ Tutorial
- ☒ Discussion
- ☒ Interaction
- ☒ Audio/Video
- ☒ Others _____

Teaching-learning Materials

- ☒ Lecture Note
- ☒ Handout
- ☒ Text & reference books
- ☒ Journal/conference papers
- ☒ Online materials
- ☒ Lab manual/lab sheet
- ☒ Others _____

Teaching and Learning Strategies (TLS)

TLS1	Delivery of lectures using multimedia of different topics
TLS2	Active and fruitful discussion in class regarding efficient solving of logical and mathematical problems
TLS3	Group discussion and presentation regarding diverse problems and corresponding lectures
TLS4	Evaluation of class performance to reach each student in a class for every topic.

a. Assessment of CO-PO, Bloom's Level, Knowledge Profile**Teaching-learning and Assessment Strategy:**

Lectures, Class Participation, Homework/Assignments, Class Tests and Semester Final Exam (Written)

Assessment Methods	Description	Marks	CO	Blooms Taxonomy
Continuous Assessment (100%)	Conduct of Lab test/ Lab Performance/ Quiz Test/Spot Test	25	CO3	C2, C3, C4, C5
	Report Writing/Programming	15	CO1, CO2, CO3	C1, C2
	Mid Term Evaluation (Exam/Project/Assignment)	20	CO1, CO2	C2, C3, C4
	Final Evaluation	30	CO1, CO2, CO3	C2, C3, C4, C5
	Viva Voce/ Presentation skills	10		
Total Marks		100	100	

Sessional Lecture Plan:

SN	Content of Course	Lecture Hours	COs	Teaching Aids	Teaching Assessment Tools
01	Sessional-01: Basic Linux Commands and practicing the Shell Scripting (Variable, input, and output)	03 Hrs.	CO1	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
02	Sessional-02: Practicing the Shell Scripting (if-else, argument passing in shell function and loop)	03 Hrs.	CO2 CO3	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
03	Sessional-03: Simulating the Process Scheduler (FCFS, SJF CPU scheduling algorithm)	03 Hrs.	CO2 CO3	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
04	Sessional-04: Simulating the Process Scheduler (Priority CPU scheduling algorithm)	03 Hrs.	CO2 CO3	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
05	Sessional-05: Simulating the Process Scheduler (Round Robin CPU scheduling algorithm)	03 Hrs.	CO2 CO3	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
06	Sessional-06: Simulating Disk Scheduling algorithm of an OS (FCFS, SSTF)	03 Hrs.	CO3 CO3	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
07	Sessional-07: Simulating Disk Scheduling algorithm of an OS (SCAN, LOOK)	03 Hrs.	CO3 CO3	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
Mid-Term Assessment (20 Marks)					
08	Sessional-08: Simulating Page Replacement Techniques (FIFO Page Replacement Algorithm)	03 Hrs.	CO1 CO2	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
09	Sessional-09: Simulating Page Replacement Techniques (LRU Page Replacement Algorithm)	03 Hrs.	CO1 CO2	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
10	Sessional-10: Simulating Page Replacement Techniques (Optimal Page Replacement)	03 Hrs.	CO2 CO3	PowerPoint slides, Text	Lectures, Class Participation/

SN	Content of Course	Lecture Hours	COs	Teaching Aids	Teaching Assessment Tools
	Algorithm)			Book, White Board	Individual or Group Discussion
11	Sessional-11: Deadlock Simulation in OS	03 Hrs.	CO3 CO3	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
12	Sessional-12: Deadlock Prevention in OS	03 Hrs.	CO3 CO3	PowerPoint slides, Text Book, White Board	Lectures, Class Participation/ Individual or Group Discussion
12	Sessional-13: Viva Voce/ Presentation skills		CO3		
14	Sessional-14 Lab final	03 Hrs.	CO1 CO2 CO3		

Minimum Attendance: 70% class attendance is mandatory for a student to appear at the final examination.

Grading System: As per the approved grading scale of Bangladesh Army University of Engineering & Technology (BAUET), Qadirabad, Natore (**Appendix-I**)

Student's responsibilities:

- Students must come to the class prepared for the course material covered in the previous class
- They must submit their assignments on time.
- No late or partial assignments will be acceptable.

Course Assessment Methods & their Weights:

Assessment Method	Marks (%)
Conduct of Lab test/Lab Performance	25
Report Writing/Programming	15
Mid Term Evaluation (Exam/Project/Assignment)	20
Final Evaluation	30
Viva Voce/ Presentation skills	10
	100

Appendix-I (Grading Policy):

Numerical Score	Letter Grade		Grade points
80% and above	A+	(A Plus)	4.00
75% to less than 80%	A	(A Regular)	3.75
70% to less than 75%	A-	(A Minus)	3.50
65% to less than 70%	B+	(B Plus)	3.25
60% to less than 65%	B	(B Regular)	3.00
55% to less than 60%	B-	(B Minus)	2.75
50% to less than 55%	C+	(C Plus)	2.50
45% to less than 50%	C	(C Regular)	2.25
40% to less than 45%	D	(D Pass)	2.00
Less than 40%	F	Fail	0.00
Assigned courses are not appeared	I	-	Incomplete
As per academic policy	W	-	Withdrawn
Failed to submit	X	-	Projects/Thesis Continuation
Due to offense	E	-	Expelled

*I-Incomplete; *W-Withdrawn; *X-Projects/Thesis Continuation; *E-Expel

*The subject in which the student gets an F grade shall not be counted towards credit hours requirements and for the calculation of Grade Point Average (GPA). Students failing in a course/ course due to being expelled, dis-collegiate, absent will be treated as F according to the Grading system.

Graduate Attributes/Program Outcomes (POs)

Program Outcomes (POs) or graduate attributes are narrower statements describing what students are expected to know and be able to do by graduation. These statements relate to the knowledge, skills, and attitudes students acquire while progressing through the program. By graduation, the program must demonstrate that students have achieved an acceptable minimum level of specific knowledge, skills, and behavioral traits. The BAETE requires that students acquire the following POs or graduate attributes (According to BAETE Accreditation Manual- III).

PO1.	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems.
PO2.	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).
PO3.	Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5)
PO4.	Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
PO5.	Create, select and apply and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems (WK2, WK6)
PO6.	When solving complex engineering problems, analyze and evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7)
PO7.	Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9)
PO8.	Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote, and distributed settings (WK9).
PO9.	Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences
PO10.	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11.	Recognize the need for, and have the preparation and ability for i) independent and lifelong learning ii) adaptability to new and emerging technologies, and iii) critical thinking in the broadest context of technological change (WK8)
PO12.	Demonstrate knowledge and understanding of the competencies necessary to transform opportunities and ideas into a new business

Complex Engineering Problems (WP) [Need High Taxonomy Level]

Range of Complex Engineering Problem Solving

Attributes	Attribute Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7
Depth of Knowledge required	WP1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first-principles analytical approach
Range of conflicting requirements	WP2: Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
Depth of analysis required	WP3: Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
Familiarity of issues	WP4: Involve infrequently encountered issues or novel problems
Extent of applicable codes	WP5: Address problems not encompassed by standards and codes of practice for professional engineering
The extent of stakeholder involvement and level of conflicting requirements	WP6: Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
Interdependence	WP7: Address high-level problems including many components or sub-problems that may require a systems approach

Complex Engineering Activities

Range of Complex Engineering Activities

Attributes	Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:
Range of resources	EA1: Involve the use of diverse resources including people, data and information, natural, financial, and physical resources, and appropriate technologies including analytical and/or design software.
Level of interaction	EA2: Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues.
Innovation	EA3: Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge.
Consequences to society and the environment	EA4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	EA5: Can extend beyond previous experiences by applying principles-based approaches

Bloom's Taxonomy Domain:

Cognitive Domain		Psychomotor Domain		Affective Domain	
C1	Remembering	P1	Perception	A1	Receive
C2	Understanding	P2	Set	A2	Respond
C3	Applying	P3	Guided Response	A3	Value
C4	Analyzing	P4	Mechanism	A4	Organize
C5	Evaluating	P5	Complex Overt Response	A5	Internalize
C6	Creating/Designing	P6	Adaption		
		P7	Origination		

Instructions for the students to conduct Sessional Class

SL	Instructions
1	The experiment list for the sessional has been displayed. Each experiment is given a number.
2	For each day, every student must have to implement the given experimental task according to the experiment number. Marks will be given by the faculty for the implemented experiment.
3	Students may be permitted to carry out missed experiments in their own time with the approval of faculty only.
4	Each sessional report must be submitted within 15 minutes after starting the class time by wrapping a cover file. The report must be in handwritten format. Each report will contain the following parts- a. Experiment No b. Name of the Experiment c. Problem Description d. Algorithm / Flowchart e. Source Code f. Input g. Output h. Discussion (This part must be unique)
5	Sessional marks awarded at the end of the semester will be strictly followed according to the sessional assessment method.

Approved by:

**Chairman, PSAC, CSE,
BAUET**



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 01

Experiment Name: Basic OS Commands in Unix/Linux

SL	Assigned Tasks	CO	BL
Q1.	File Commands: 1. ls Directory listing 2. ls -al Formatted listing with hidden files 3. ls -lt Sorting the Formatted listing by time modification 4. cd dir Change directory to dir 5. cd Change to home directory 6. pwd Show current working directory 7. mkdir dir Creating a directory dir 8. cat >file Places the standard input into the file 9. more file Output the contents of the file 10. head file Output the first 10 lines of the file 11. tail file Output the last 10 lines of the file 12. tail -f file Output the contents of file as it grows, starting with the last 10 lines 13. touch file Create or update file 14. rm file Deleting the file 15. rm -r dir Deleting the directory Process management: 1. ps To display the currently working processes 2. top Display all running process Unix/Linux Command Reference 3. kill pid Kill the process with given pid 4. killall proc Kill all the process named proc 5. pkill pattern Will kill all processes matching the pattern System Info: 1. date Show the current date and time 2. cal Show this month's calender 3. uptime Show current uptime 4. w Display who is on line 5. whoami Who you are logged in as	CO1	C1, C2



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 02

Experiment Name: Practices on Operating System Security

SL	Assigned Tasks	CO	BL
Q1.	Get familiar with the following commands: 1. chmod - modify file access rights 2. su - temporarily become the superuser 3. sudo - temporarily become the superuser 4. chown - change file ownership 5. chgrp - change a file's group ownership Compression: 1. tar cf file.tar file Create tar named file.tar containing file 2. tar xf file.tar Extract the files from file.tar 3. tar czf file.tar.gz files Create a tar with Gzip compression 4. tar xzf file.tar.gz Extract a tar using Gzip 5. tar cjf file.tar.bz2 Create tar with Bzip2 compression Network Related: 1. ping host Ping host and output results 2. whois domain Get whois information for domains 3. dig domain Get DNS information for domain 4. dig -x host Reverse lookup host Shortcuts: 1. ctrl+c Halts the current command 2. ctrl+z Stops the current command, resume with fg in the foreground or bg in the background 3. ctrl+d Logout the current session, similar to exit 4. ctrl+w Erases one word in the current line 5. ctrl+u Erases the whole line	CO1 CO2	C1,C2, C3, C4



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 03

Experiment Name: Practicing on Process Scheduler (Part-1)

SL	Assigned Tasks	CO	BL
Q1.	First Come First Serve Process Scheduling: For FCFS scheduling algorithm, you shall get the number of processes/jobs in the system and their CPU burst times. This is a non-preemptive scheduling where a process getting the CPU will finish its works and the leave the CPU after it exhausts its required CPU time. The scheduling is performed on the basis of arrival time of the processes irrespective of their other parameters. Each process will be executed according to its arrival time. The waiting time and turnaround time of each of the processes can be calculated as well as the average system turnaround and waiting time. Example: Process Arrival Time CPU Time P1 4 5 P2 0 7 P3 2 9 Gantt chart: <pre> -----P2----- -----P3----- -----P1----- 0 7 16 21 </pre> Shortest Job First (Non-Preemptive version) For the Shortest Job First scheduling algorithm, you shall get the number of processes/jobs in the system and their CPU burst times. This is a non-preemptive scheduling where a process getting the CPU will finish its works and the leave the CPU after it exhausts its required CPU time. The scheduling is performed on the basis of the shortest CPU time of the processes irrespective of their other parameters. Each process will be executed according to its arrival time. The waiting time and turnaround time of each of the processes can be calculated as well as the average system turnaround and waiting time. Example Process Arrival Time CPU Time P1 4 5 P2 0 7 P3 2 9 Gantt chart: <pre> -----P2----- -----P1----- -----P3----- 0 7 12 21 </pre>	CO2 CO3	C3, C4



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 04

Experiment Name: Practicing on Process Scheduler (Part-2)

SL	Assigned Tasks	CO	BL
Q1.	Round Robin Scheduling Algorithm A queue of processes is maintained for all the available processes in the system according to their arrival time. A process P in the system is selected as per the FCFS rule and given opportunity in the CPU for a pre-defined time quantum Q. If the remaining CPU time r of the process is less than the time quantum Q, a context switch takes place after r amount of time and the process P leaves the queue. Otherwise, the process P runs for Q time quantum and a context switch takes place after Q amount of time. The process P goes to the end of queue for more time quantum until it finishes its required amount of time. Example Process Arrival Time CPU Time P1 4 5 P2 0 7 P3 6 9 P4 10 9 Time Quantum Q = 3 Gantt chart: <pre> ---P2--- ---P2--- ---P1--- ---P3--- ---P2--- ---P1--- ---P4--- ---P3--- ---P4--- ---P3--- ---P4--- 0 3 6 9 12 13 15 18 21 24 27 30 </pre> Priority Scheduling Priority scheduling is one of the most common scheduling algorithms in batch systems. Each process is assigned a priority. Process with the highest priority is to be executed first and so on. Processes with the same priority are executed on first come first served basis. If used as a pre-emptive version, a running process having lower priority will be sent back to the ready queue if a process with higher priority arrives in the queue. Process Arrival Time CPU Time Priority P1 4 5 0 P2 0 7 2 P3 2 9 115 Gantt chart: (Non Pre-emptive Version) <pre> -----P2----- -----P1----- -----P3----- 0 7 12 21 </pre> Gantt chart: (Pre-emptive Version) <pre> -----P2----- -----P3----- -----P1----- -----P3----- ----- P2----- 0 2 4 9 16 21 </pre>	CO3 CO3	C3, C4



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 05

Experiment Name: Implementation of threads and how to use them in Java and Simulating the Producer-Consumer Problem

SL	Assigned Tasks	CO	BL
Q1.	Producer-consumer problem also known Ananya Sarker Assistant Professor & Head Dept. of CSE, BAUET bounded-buffer problem. The problem describes two processes, the producer and the consumer which share a common fixed-size buffer used Ananya Sarker Assistant Professor & Head Dept. of CSE, BAUET a queue. • A producer's job is to generate data, putting it into the buffer and start again. • The purpose of the consumer is to consume the available data from the buffer and empty the buffer. The standard solution to this problem uses three semaphores: empty and full, which count the number of empty and full slots in the buffer, and mutex, which is a binary (or mutual exclusive) semaphore that protects the actual insertion or removal of items in the buffer. The producer and consumer are running as separate threads and they will move items to and from a buffer that is synchronized with the empty and full data structures. Input: Producer 1: Data to insert in the buffer Data: 10 Producer 3: Data to insert in the buffer Data: 110 Output : Producer 1: Data 10 is inserted in the buffer Empty space : n-1 Full space : 1 Producer 3: Data 110 is inserted in the buffer Empty space : n-2 Full space : 2 Consumer 0: Consumed Data 10 from the buffer Full space : 1 Consumer 3: Consumed Data 110 from the buffer Full space : 0	CO3 CO3	C3, C4, A4



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3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 06

Experiment Name: Simulating the Reader-Writer Problem

SL	Assigned Tasks	CO	BL
Q1.	There is a shared resource which should be accessed by multiple processes. There are two types of processes in this context. They are reader and writer. Any number of readers can read from the shared resource simultaneously, but only one writer can write to the shared resource. When a writer is writing data to the resource, no other process can access the resource. A writer cannot write to the resource if there are non-zero number of readers accessing the resource at that time. From the above problem statement, it is evident that readers have higher priority than writer. If a writer wants to write to the resource, it must wait until there are no readers currently accessing that resource. You need to implement the code in Java programming language. Pseudocode In the standard solution, we use one mutex m and a semaphore w. An integer variable read_count is used to maintain the number of readers currently accessing the resource. The variable read_count is initialized to 0. A value of 1 is given initially to m and w. Instead of having the process to acquire lock on the shared resource, we use the mutex m to make the process to acquire and release lock whenever it is updating the read_count variable. Input / Output: Data in Buffer : 1 5 6 7 8 9 15 Reader 1: Reading 7 Reader 2: Reading 1 Reader 3: Reading 15 Reader 3: Leaves Reader 2: Leaves Reader 1: Leaves Writer 0 : Writes in the system 23, 45 Data in Buffer : 5 6 8 9 23 45 Writer 1 : Waiting Reader 1: Waiting Reader 2: Waiting Writer 0 : Leaves the system Writer 1 : Writes in the system 3, 4 Data in Buffer : 5 6 8 9 23 45 3 4 Writer 1 : Leaves the system Reader 1: Reading 5	CO1 CO2	C1,C2, C3, C4



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3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 07

Experiment Name: Deadlock detection using resource allocation graph for a single quantity of each resources using a cycle detection algorithm

SL	Assigned Tasks	CO	BL
Q1.	Deadlock: From the application point of view, a deadlock is a situation in which two computer programs are sharing the same resource and effectively preventing each other from accessing the resource, resulting in both programs ceasing to function. Deadlock detection using Cycle detection method Algorithm: 1. For each node, N in the graph, perform the following five steps with N as the starting node. 2. Initialize L to the empty list, designate all arcs as unmarked. 3. Add current node to end of L, check to see if node now appears in L two times. If it does, graph contains a cycle (listed in L), algorithm terminates. 4. From given node, see if any unmarked outgoing arcs. If so, go to step 5; if not, go to step 6. 5. Pick an unmarked outgoing arc at random and mark it. Then follow it to the new current node and go to step 3. 6. If this is initial node, graph does not contain any cycles, algorithm terminates. Otherwise, dead end. Remove it, go back to previous node, make that one current node, go to step 3. Input: A Directed graph (nodes can be either resource or process) Output: Number of cycles / number of deadlocks existing Input: Number of nodes : 13 Node names : R A C S D T B E F U V W G Edges: R to A A to S C to S Output 1 - Deadlock Detected Among nodes: D T E V G U Additional Task: Recovery from a Deadlock When you are taking input of the node names, you need to ask whether the node is a resource or a process. You can recover deadlock in the following way, 1. Pre-empt a resource from a process in the existing cycle and allocate it to another process to break the deadlock 2. Kill a process and give its resources to the process looking for it. Brainstorm to derive a solution code for this task	CO2 CO3	C3, C4



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Full Marks: 20

Time: 03 Hours

Experiment No: 08

Experiment Name: Simulate Banker's Algorithm for Deadlock Avoidance

SL	Assigned Tasks	CO	BL																
Q1.	In Banker's algorithm, when a new process enters a system, it declares the maximum number of instances of each resource type it needed. This number may exceed the total number of resources currently available in the system. When a process requests a set of resources, the system must determine whether the allocation of each resource will leave the system in safe state. If it will, the resources are allocated; otherwise, the process must wait until some other process release the resources. If all the processes in the system can be satisfied in any order by giving them their requested number of resources, the system is said to be in safe state and the order of serving is called the safe sequence. Example: Resource X Y Z X Y Z Resources= {20, 19, 16} Allocation Max Need <P2, P1, P6, P7, P3, P4, P5>is the safe sequence <table><tr><th>X Y Z</th><th>X Y Z X Y Z</th></tr><tr><td>P1 1 1 1</td><td>7 5 6 6 4 5</td></tr><tr><td>P2 2 3 3</td><td>5 8 5 3 5 2</td></tr><tr><td>P3 3 0 5</td><td>9 2 8 6 2 3</td></tr><tr><td>P4 2 2 0</td><td>8 8 6 6 6 6</td></tr><tr><td>P5 3 1 0</td><td>14 4 5 11 3 5</td></tr><tr><td>P6 1 2 1</td><td>4 6 3 3 4 2</td></tr><tr><td>P7 1 1 1</td><td>2 9 2 1 8 1</td></tr></table>	X Y Z	X Y Z X Y Z	P1 1 1 1	7 5 6 6 4 5	P2 2 3 3	5 8 5 3 5 2	P3 3 0 5	9 2 8 6 2 3	P4 2 2 0	8 8 6 6 6 6	P5 3 1 0	14 4 5 11 3 5	P6 1 2 1	4 6 3 3 4 2	P7 1 1 1	2 9 2 1 8 1	CO3 CO3	C3, C4, A4
X Y Z	X Y Z X Y Z																		
P1 1 1 1	7 5 6 6 4 5																		
P2 2 3 3	5 8 5 3 5 2																		
P3 3 0 5	9 2 8 6 2 3																		
P4 2 2 0	8 8 6 6 6 6																		
P5 3 1 0	14 4 5 11 3 5																		
P6 1 2 1	4 6 3 3 4 2																		
P7 1 1 1	2 9 2 1 8 1																		

	Program Input: Enter the no. of processes: 4Enter the no. of resources:3 Process 1 Maximum value for resource 1:3Maximum value for resource 2:2Maximum value for resource 3:2Allocated from resource 1:1 Allocated from resource 2:0 Allocated from resource 3:0 Process 2 Maximum value for resource 1:6Maximum value for resource 2:1Maximum value for resource 3:3Allocated from resource 1:5 Allocated from resource 2:1 Allocated from resource 3:1 Process 3 Maximum value for resource 2:1Maximum value for resource 3:4Allocated from resource 1:2 Allocated from resource 2:1 Allocated from resource 3:1 Process 4 Maximum value for resource 1:4Maximum value for resource 2:2Maximum value for resource 3:2Allocated from resource 1:0 Allocated from resource 2:0 Allocated from resource 3:2 Enter total value of resource 1:9 Enter total value of resource 2:3Enter total value of resource 3:6	Program Output: The System is currently in safe state and < P2 P1 P3 P4 > is the safe sequence.	
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Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 09

Experiment Name: Simulating Memory Management of an OS

SL	Assigned Tasks	CO	BL												
Q1.	Implementing the general concepts of the following Memory allocation techniques and fragmentations. such as: First-fit, Best Fit, Worst-fit (In Fixed/ variable length partitioning). Example: How to satisfy a request of size n from a list of free holes First-fit: Allocate the first hole that is big enough to fit the request. Best-fit: Allocate the smallest hole that is big enough; must search entire list, (unordered list by size) Worst-fit: Allocate the largest hole; must also search entire list. <table><tr><th>Memory Block</th><th>Size</th></tr><tr><td>Block 1</td><td>50</td></tr><tr><td>Block 2</td><td>200</td></tr><tr><td>Block 3</td><td>70</td></tr><tr><td>Block 4</td><td>115</td></tr><tr><td>Block 5</td><td>15</td></tr></table> Memory Requests: 100, 10, 35, 15, 23, 6, 25, 55, 88, 40 Program: Input: Memory Holes: 50 200 70 115 15 Memory Requests: 100, 10, 35, 15, 23, 6, 25, 55, 88, 40 Output: First Fit Memory Allocation Step by step (See the example above) External Fragmentation: 93	Memory Block	Size	Block 1	50	Block 2	200	Block 3	70	Block 4	115	Block 5	15	CO3 CO3	C3, C4, A4
Memory Block	Size														
Block 1	50														
Block 2	200														
Block 3	70														
Block 4	115														
Block 5	15														



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 10

Experiment Name: Simulating Page Replacement techniques of an OS (Part-1)

SL	Assigned Tasks	CO	BL
Q1.	FIFO Page Replacement Algorithm: If there is no space left in memory, select the page came earlier in the memory compared to the available pages in the memory and replace that page with the new one. Optimal Page Replacement Algorithm: If there is no space left in memory, select the page that will either not be used or used later compared to the available pages in the memory and replace that page with the new one. Least Recently Used Algorithm: If there is no space left in memory, select the page that has not been used recently compared to the available pages in the memory and replace that page with the new one. A page which is not used recently will be replaced. Example Reference string: 7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1 3 frames (3 pages can be in memory at a time per process) Input: Number of pages: 8 Number of Page References: 22 Reference String: 7,0, 1 , 2, 0 , 3 ,0 , 4 , 2 ,3 , 0 ,3 , 0 , 3 ,2 , 1 ,2 , 0 , 1 ,7 , 0 Number of Memory Page Frame: 3 Output: Number of page fault using FIFO Page replacement Algorithm: Page Fault Rate: 68% Number of page fault using Optimal Page replacement Algorithm: Page Fault Rate: 41% Number of page fault using Least Recently Used Page replacement Page Fault Rate: 55%	CO1 CO2	C1, C2, C3, C4



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 11

Experiment Name: Simulating Page Replacement techniques of an OS (Part-2)

SL	Assigned Tasks	CO	BL
Q1.	Least Recently Used page replacement algorithm using stack If there is no space left in memory, select the page that has not been used recently compared to the available pages in the memory and replace that page with the new one. A page which is not used recently will be replaced. Input: Number of pages: 8 Reference String: 7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1 Number of Memory Page Frame: 3 Output: <pre> 7 0 1 2 0 3 0 4 2 3 0 3 0 3 2 1 2 0 1 7 0 1 7 0 1 2 0 3 0 4 2 3 0 3 0 3 2 1 2 0 1 7 0 7 0 1 2 2 3 0 4 2 2 2 2 0 3 3 1 2 0 1 7 </pre> Enhanced second chance algorithm Uses Reference bit and Modify bit to select a page for replacement. (0, 0) – Neither recently used nor modified – best page to replace. (0, 1) – Not recently used but modified – not Ananya Sarker Assistant Professor & Head Dept. of CSE, BAUET - good because we need to swapout a page, but still better than used pages. (1, 0) – Recently used but unmodified. (1, 1) – Recently used and modified – the worst page to replace.	CO2 CO3	C3, C4



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 12

Experiment Name: Practicing on disk-Arm Scheduling Algorithm

SL	Assigned Tasks	CO	BL
Q1.	Disk Arm Scheduling: The operating system tries to use hardware efficiently for disk drives access time, disk bandwidth Access time has two major components: • Seek time is time to move the heads to the cylinder containing the desired sector • Rotational latency is additional time waiting to rotate the desired sector to the disk head. FCFS (First Come First Serve) Scheduling: The cylinder requests are served in the same order they appear in the request pool. We illustrate them with a request queue (0-199 cylinder). Cylinder Requests: 98, 183, 37, 122, 14, 124, 65, 67 ; Head pointer starts at 53 SSTF (Shortest Seek time First) Scheduling The next cylinder request is served which is nearby from the current serving location in the request pool. We illustrate them with a request queue (0-199 cylinder). Cylinder Requests: 98, 183, 37, 122, 14, 124, 65, 67; Head pointer starts at 53 C-SCAN Algorithm In this algorithm, the service is provided only in one direction and the head moves from the first cylinder to the last cylinder and serves the request only in one direction. Input: (FOR FCFS) Number of heads: 200 Cylinder requests: 7 0 1 2 0 3 0 4 2 3 0 3 0 3 2 1 2 0 1 7 0 1 Current head position: 53 Output: Cylinder Serving Order: 53 -> 98 -> 183 -> 37 -> 122 -> 14 -> 124 -> 65 -> 67 Total Cylinder movement: 640	CO3 CO3	C3, C4, A4



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

3rd Year 1st Semester

Course Code: CSE-3106

Course Title: Operating Systems Sessional

Full Marks: 20

Time: 03 Hours

Experiment No: 13

Experiment Name: Practicing on File Management system

SL	Assigned Tasks	CO	BL
Q1.	Contiguous allocation (Sequential): It requires each file to occupy a set of contiguous blocks on the hard disk where disk address define a linear ordering on the disk. Algorithm • Create a one-dimension array of size n (n is the number of blocks in the disk) Read a filename from the user that is about to be created and allocated blocks Read the file-size (S - number of blocks) • Now, Create a unique number for the file. • Read your One-dimension array to look for a free space of S number of contiguous blocks. • If available, allocate the file those blocks and mark in your array Continue to read q number of files in the same way • If, space cannot be allocated, inform the user that, the file could not be created Now, read a file name from the user and • show its occupied blocks or 'NA' if the file does not exist. Input/Output: Enter total number of blocks: 200 Enter Filename: A Enter File-Size: 4 blocks File A created Enter Filename: B Enter File-Size: 400 blocks File B cannot be created (not enough free blocks) Enter Filename: C Enter File-Size: 40 blocks File C created Search Filename: A File Found in the blocks: 5, 6, 7, 8 Search Filename: D File not Found. Non-Contiguous allocation: A file may be allocated any free blocks which may not be contiguous but free.	CO3 CO3	C3, C4, A4

	Algorithm • Create a one-dimension array of size n (n is the number of blocks in the disk) Read a filename from the user that is about to be created and allocated blocks Read the file-size (S - number of blocks) • Now, Create a unique number for the file. • Read your one-dimension array to look for a free space of S number of contiguous or non-contiguous blocks. • If available, allocate the file those blocks and mark in your array Continue to read q number of files in the same way • If, space cannot be allocated, inform the user that, the file could not be created • Now, read a file name from the user and show its occupied blocks or 'NA' if the file does not exist. Input/Output: Enter total number of blocks: 200 Enter Filename: A Enter File-Size: 4 blocks File A created Enter Filename: B Enter File-Size: 400 blocks File B cannot be created (not enough free blocks) Enter Filename: C Enter File-Size: 40 blocks File C created Search Filename: A File Found in the blocks: 1, 9, 10, 20 Search Filename: D File not Found.		
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*****End*****